

1, 2, 3 or 4 for given i, t

$\text{choice}_{it} \sim \text{categorical-logit}(Q_{\text{sum}})$

$$Q_{\text{sum}} = Q_r + Q_p$$

for $t=1$, $Q_r = Q_p = 0$ (init)

for $t > 1$,

$$PE_r = \underbrace{R_i}_{\text{reward value}} \times \underbrace{\text{rew}_{it}}_{\text{reward}} - \underbrace{Q_r}_{\text{reward expectation for chosen arm}}_{i, t, \text{choice}}$$

prediction error reward for chosen arm
reward sensitivity
how much you like reward!

$$PE_p = P_i \times \text{pun}_{it} - Q_p_{i, t, \text{choice}}$$

~~Q_r = 0~~ FICTIVE UPDATING OF UNCHOSEN OPTIONS

$$PE_r = 0 - Q_r$$

$$PE_p = 0 - Q_p$$

No reward / punishment

vector of punishment expect

For chosen options:

$$Q_{r_{\bar{i}}, t, \text{chosen}} = Q_{r_{\bar{i}}, t, \text{chosen}} + \alpha_{r_{\bar{i}}} PE_r$$

$$Q_{p_{\bar{i}}, t, \text{chosen}} = Q_{p_{\bar{i}}, t, \text{chosen}} + \alpha_{p_{\bar{i}}} PE_p$$

For unselected options:

$$Q_{r_{\bar{i}}, t, \text{unchosen}} = Q_{r_{\bar{i}}, t, \text{unchosen}} + \alpha_{r_{\bar{i}}} PE_{\bar{r}}$$

$$Q_{p_{\bar{i}}, t, \text{unchosen}} = Q_{p_{\bar{i}}, t, \text{unchosen}} + \alpha_{p_{\bar{i}}} PE_{\bar{r}}$$